FINAL

Title

Volition: Adaptive Biosignal Platform for Universal Assistive Control

Problem (450 chars)

As per Census 2011, 54 lakh Indians live with motor disabilities - conservatively over 65 lakh today. Assistive devices like myoelectric arms and smart wheelchairs cost ₹1- 8 lakh, require professional calibration, and depend on internet connectivity. For 95% of Indian families, these aren't just expensive; they're completely out of reach. No affordable, offline, and self-calibrating easy-to-use solution currently exists.

Objectives (500 chars)

To develop an offline adaptive biosignal platform that reads EMG, EOG, and EEG signals to map each user's neural signature (targeting >80% accuracy) through a one-time guided setup - without clinical calibration or internet - and translate it into actionable commands for computers, household appliances, and mobility aids via one universal application. Ultimately, the objective is to make assistive technology accessible and affordable to every affected Indian, regardless of income or location.

Methods (500 chars)

The NPG Lite captures biopotential signals from skin-contact electrodes and streams them wirelessly through Bluetooth(BLE) to the Volition application. A lightweight machine learning pipeline processes these streams in parallel to classify intents from different body parts. The software’s architecture is engineered to map these classified intents directly into digital control commands, routing them wirelessly to connected devices and hardware interfaces.

Scientific Approach (350)

Volition fuses multi-modal biosignals with real-time sensing. EMG (20-500Hz) tracks muscle twitches, EOG captures eye movement, and EEG beta waves (12-30Hz) reflect cognitive intent. A background Passive Floor Tracker auto-calibrates resting baselines, eliminating single-channel false triggers for reliable assistive control.

Results (400)

Biopotentials are transmitted successfully over BLE and classified into EMG, EOG, and EEG intents through self-calibrating thresholds requiring only a brief guided setup, driving real-time computer control offline. This is achieved using reusable, moisture-activated semidry electrodes in place of disposable ones, keeping per-use costs negligible for long-term, everyday use.

Conclusions (400)

By offering a ₹4,000 - 8,500 alternative to ₹1-8 lakh commercial assistive devices, Volition brings physical independence within reach of India's 65 lakh+ motor-impaired citizens, with a modular, open-source architecture designed for future expansion.

Target Population(100)

Everyday Indians reclaiming their physical autonomy and dignity after severe motor impairments.

Social Impact(200)

After a one-time setup at any of India's 714 government district hospitals, Volition runs entirely offline & self-calibrates at no cost, restoring autonomy and easing caregiving burdens nationwide.

Bibliography(200)

Census of India 2011, Disabled Persons in India, censusindia.gov.in
UDL NPG Lite Schematics, github.com/upsidedownlabs
TensorFlow Lite, tensorflow.org/lite
Health Dynamics of India 2022-23, MoHFW